Modifying LeanCoP Options On the Fly

(aka "the PAAR paper")

Introduction and Research

LeanCoP inspired a large family of provers which, with a small amount of modifications, could improve the performance of the connection prover in a large set of problems[2]. The ones that inspired this work change one property of the original LeanCoP at the initial time.

LeanCoP 2.0[1] added several options, most notably `cut` and `scut` to restrict backtracking in connection calculus. That version also enables *iterative deepening* with `comp(J)`, which iteratively limits the height 1 at a time of the tree when the prover is saturated[1] until it reaches maximum height `J`. Afterwards it restarts iterative deepening from 1, but without any cut active.

Novel Idea

Implementation

Preliminary Results

References

[1] Jens Otten. Restricting backtracking in connection calculi. 23(2–3):159–182, 2010.

[2] Jens Otten. 20 years of leanCoP — an overview of the provers. In *Proceedings of the 1st International Workshop on Automated Reasoning with Connection Calculi (AReCCa@TABLEAUX 2023)*, volume 3613 of *CEUR Workshop Proceedings*, 2023.